

Discrete Mathematics

Moritz

Lesson 1: Propositional Logic

2026-10-01

We introduce the basic operators for boolean logic. Let P and Q be propositions.

Definition (Conjunction): The conjunction of P and Q , denoted $P \wedge Q$, is true if and only if both P and Q are true.

Theorem (De Morgan's Laws): For any two propositions P and Q , the following equivalences hold:

$$\begin{aligned}\neg(P \wedge Q) &\equiv \neg P \vee \neg Q \\ \neg(P \vee Q) &\equiv \neg P \wedge \neg Q\end{aligned}\tag{1}$$

Homework Assignment

Due: 2026-10-08

1. Complete exercises 1.1 through 1.5.
2. Prove De Morgan's laws using complete truth tables.

Lesson 2: Set Theory

2026-08-31

Today we shift focus to collections of distinct objects... and i love you